

The “Issue Tracker” MVP

Objective

Build a minimal but production-minded Issue Tracker that supports users, projects, and issues, with real-time updates and role-based access.

1. Write this app using Angular 20 & Java (JDK 17).
2. Questions? State any assumptions you make and move forward.
3. Return your deliverable within 6 hours.

The exercise evaluates your ability to design and reason about a full-stack application of realistic scope. It is intentionally sized to require thoughtful use of AI, reflecting how modern engineering teams work. However, the use of AI is not a requirement.

A core objective is to deliver a **complete, working solution** within the assignment timeframe. A project that is 100% functional across a narrower scope is preferred over a broad, unfinished architecture.

Your submission should demonstrate both full-stack capability and your ability to balance timeliness with pragmatism, using AI as a force multiplier when appropriate, and grounded in strong engineering fundamentals.

Scope (MVP):

- **Auth:**
 - Sign up / login (email+password) or OAuth

- Protected routes
- **Domain:**
 - Project with name, owner
 - Issue with title, description, status (open, in-progress, closed), priority
 - Issues assigned to users and owned by projects
- **API:**
 - CRUD for Projects and Issues
 - Server-side pagination & filtering (`status`, `priority`, `assigneeId`, `tag`, text search on `title`).
- **UI:**
 - Project list & detail.
 - Issue list with filters, sort, and text search.
 - Issue detail view with edit, comment thread, and activity log.
- **Real-time:** Push updates when an issue changes (WebSocket/SSE).
- **Data model:** Relational or document—justify the choice and indexing strategy.
- **Quality gates:**
 - Unit tests for core business logic.
 - At least one integration test hitting API + DB.
 - Basic input validation and error handling.
- **Ops hygiene:**
 - Minimal README: stack, run instructions, trade-offs, and “If I had 2 more days” section.
 - Turnkey startup: `Dockerfile` + `docker-compose` (or equivalent) to run API, DB, and web app; any required seed data